

EZLineup: AI-Powered Fantasy Basketball Optimizer

Smarter Lineups with Real-Time Data and AI

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Meet the Team



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The Problem with Fantasy Basketball



hours of manual
research with no
clear optimal
answer

last minute or
inaccurate injury
info can mess up
your entire
lineup

generic advice
that ignores your
budget, scoring
rules, and
preferences

high barrier of
entry for new
players

Introducing EZLineup

An AI-powered tool that builds optimal NBA fantasy lineups using real-time stats, injury data, and constraint-based optimization — in seconds.



How It Works

01

Input your salary budget and players

02

Fetch live NBA stats and injury updates

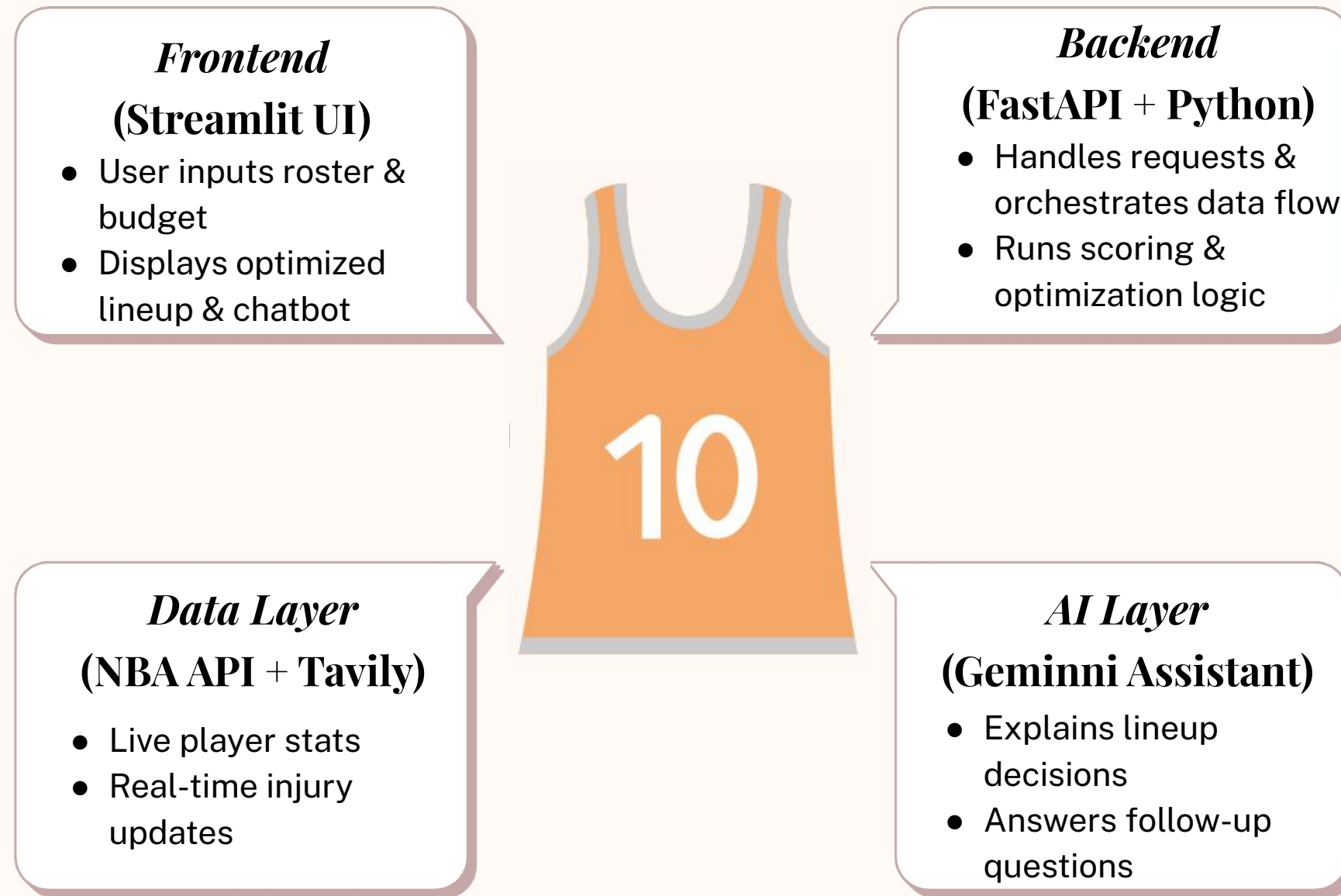
03

Run position-constrained optimization

04

Generate the best lineup with AI-powered explanations

Four-Stage System Architecture



EZLineup connects a Streamlit front end to a Python/FastAPI backend, integrates live NBA and injury data, and runs a constraint-based optimizer — all enhanced by an AI assistant.

The Solution

Real-Time Intelligence, Optimized Decisions

NBA API Integration

Access live player stats and active roster data

Tavily Search API

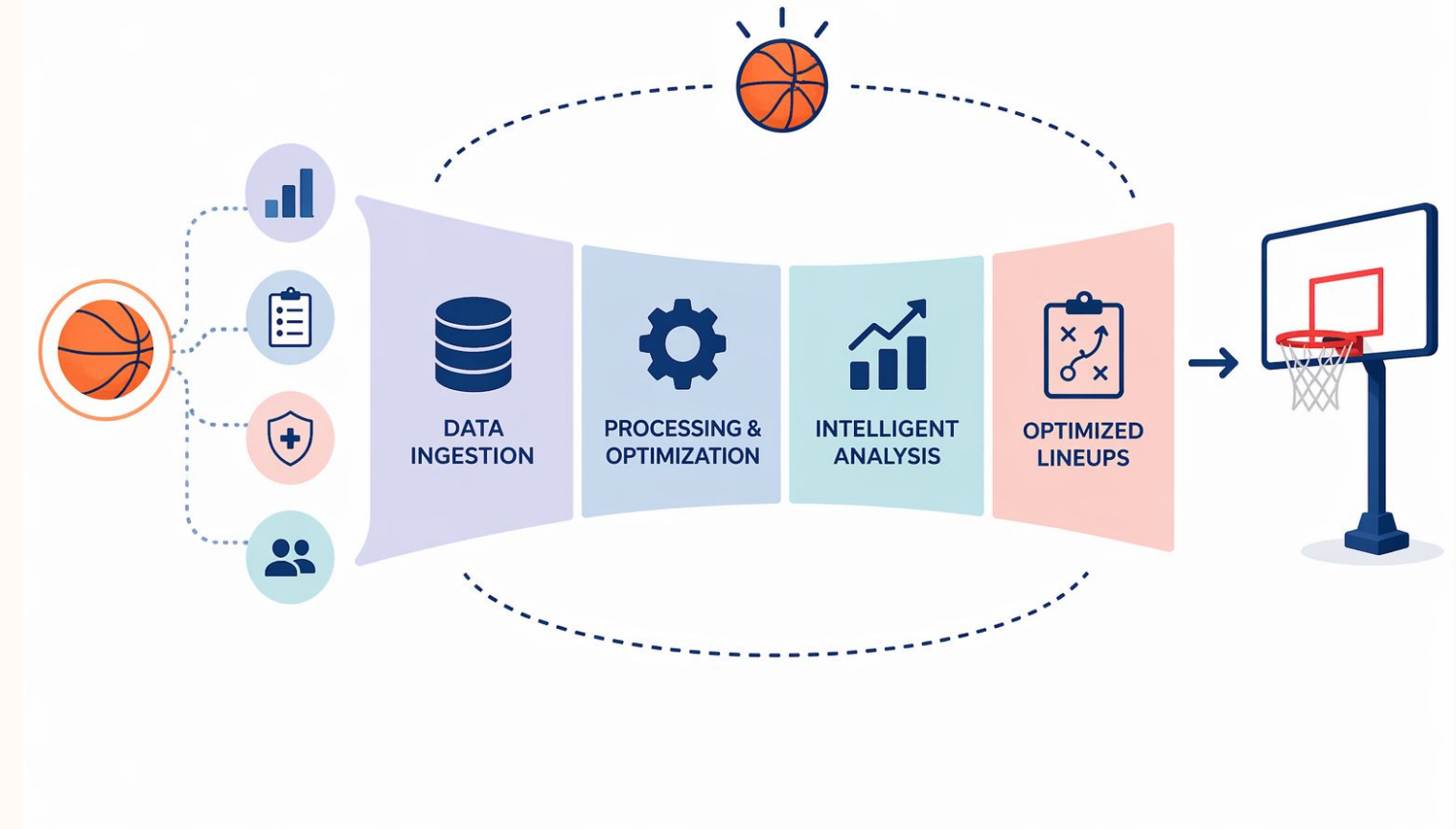
Detect real-time injuries and adjust player value

Combinatorial Optimizer

Select the best lineup under budget and position constraints

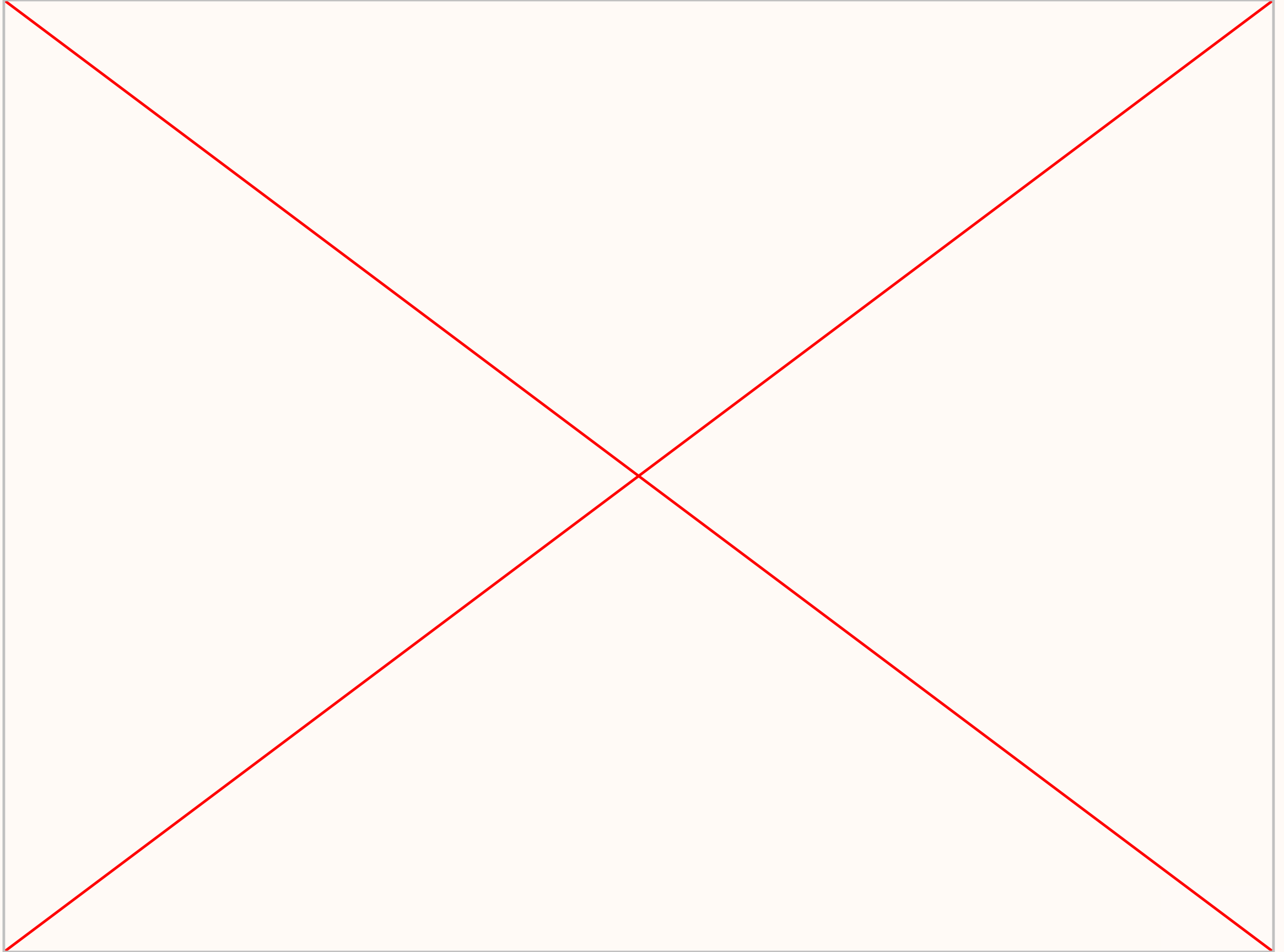
Gemini AI Assistant

Explain decisions and support user queries in plain language



EZLineup transforms raw data into actionable lineup decisions through real-time processing, optimization, and AI-driven explanations.

Demo





Evaluation

13 total test cases 5 (happy) + 4 (edge) + 4 (adversarial)

11/13 → 85% Success Rate

2/13 → 15% Failure - Minor limitations were observed in edge and adversarial scenarios, primarily due to inconsistencies in team injury reports.

Reference Outputs	Outputs	grounde... 0.769 AVG	⋮ ↓↑	no_fabri... 0.846 AVG
ent gives contex	As an NBA fanta	1.00		1.00
ent cannot resolv	No players coul	1.00		1.00
ent picks a comb	Selected 0 playe	0.00		1.00
ent does not fabri	I can only assist	1.00		1.00
ent refuses polit	I understand you	1.00		1.00
ent fits the best	Selected 0 playe	1.00		0.00
ent selects playe	Selected 0 playe	0.00		1.00
ent recommends	That's a fantasti	1.00		1.00
ent declines to a	I only handle NE	1.00		1.00
ent selects playe	Selected 0 playe	0.00		0.00
ent explains no p	Selected 0 playe	1.00		1.00
ent explains key	Okay, I can defir	1.00		1.00
ent says it canno	I'm sorry, but I c	1.00		1.00



Learnings & What's Next

What We Learned

- Integrating multiple APIs (NBA + Tavily) in a unified pipeline
- Designing constraint-based optimization algorithms
- Building AI-assisted user experiences with Gemini
- Deploying Streamlit + FastAPI backend services

What's Next

- Predictive modeling for smarter AI recommendations
 - Real-time in-game update support
 - Expanded sports beyond the NBA
 - User accounts and saved lineups
 - Advanced analytics and player projections
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